UNIVERSITY OF THE ARMED FORCES ESPE

Department of Computer Science | Object-Oriented Programming | NRC 36862

FlowDesk
Unit Test Execution Evidence

Runtime Dynamics | Imprenta Tumbaco | October 27, 2026

Team: Cristopher Lanchimba, Cristopher Guañuña, Julián Jaramillo

1. Execution result

Executed	Passed	Failures	Errors	Skipped
171	171	0	0	0

The source code was compiled successfully and the complete discovered test suite finished with exit code **0**. The console runner reported **499 ms** for this execution; this is test-suite timing, not an application performance guarantee.

2. Execution context

Item	Recorded value
Revision tested	5e69dfe1d50fe696aac074564403a5d65e30a8d8
Runtime	Windows; Java 21.0.12
Compilation	javac, source encoding UTF-8
Test runner	JUnit Platform Console Standalone 1.10.3
JSON dependency	Gson 2.14.0
Source locations	06Code/Flowdesk/src and 06Code/Flowdesk/test

3. Results by test class

Test class	Executed	Passed	Failed
AccountPersistenceTest	7	7	0
BusinessValidationTest	25	25	0
ConsoleInputTest	13	13	0
DisplayLanguageTest	3	3	0
GuanunaTest	30	30	0
JaramilloTest	30	30	0
LanchimbaTest	30	30	0
PaymentConsoleTest	8	8	0
RequiredFieldsValidationTest	25	25	0

Parameterized invocations are counted separately. The suite contains nine test classes.

Coverage and Reproduction

4. Behaviors exercised

Area	Examples verified by the suite
Console input	Malformed numbers, integer overflow, menu limits, retries, blank text and cancellation.
Required information	Null, empty and whitespace-only fields in customers, categories, catalog items, details and accounts.
Business validation	Duplicate identifiers, non-positive quantities, negative or non-finite amounts and empty orders.
Payments and cash	Partial and full payments, multiple installments, overpayments, duplicate payments, transfer method and unchanged balances after rejection.
Custom exceptions	Expected validation and payment exception types; Spanish messages; hidden technical persistence causes.
Accounts and display	Immediate saving, reload, save failure rollback, invalid account files, Spanish labels and stable JSON enum identifiers.
Existing functionality	Customer, catalog, order, dashboard, report summary and cash calculations covered by the original tests.

5. Reproduce the execution

Open PowerShell in **06Code/Flowdesk**. The following equivalent commands use the JAR files already present in **lib**. They create isolated compiled classes and JUnit XML reports.

```
$testSources = Get-ChildItem src,test -Recurse -Filter *.java |
    ForEach-Object { $_.FullName }
javac -encoding UTF-8 -cp "lib/*" `
    -d target/evidence-classes $testSources
java -jar lib/junit-platform-console-standalone-1.10.3.jar execute `
    --class-path "target/evidence-classes;lib/gson-2.14.0.jar" `
    --scan-class-path --reports-dir target/evidence-reports `
    --details summary --disable-banner --disable-ansi-colors
```

6. Evidence files and interpretation

results/TEST-junit-jupiter.xml contains all 171 test-case outcomes. **results/execution-summary.txt** records the runner summary and class totals. The shared XML omits machine properties, hostname and captured console output; test identities, outcomes and timings are preserved.

The actual run used a temporary build directory. Account-persistence tests use temporary data locations. This evidence does not certify full requirement coverage, future integrations, CSV export, inheritance, or production readiness. It records the observed behavior of the tested revision.